

SV318

SPECTR

Kathleen Peel takes the lid off the first MSX standard computer — the Spectravideo SV318 which should be able to run the same software as that used by the next wave of Japanese micros.



Built-in joystick lifts out so that base can be used as four-directional keypad to control cursor.

AVIDEO

THE REVIEW MODEL, ONE of the few MSX computers in this country, is a Spectravideo 318 and has been kindly loaned to me by Artic Computing which is at present translating the excellent Spectrum Chess II and selected other games from its range into MSX format.

The Spectravideo is not, in fact, a UK version, but an American machine that operates on the NTSC television system. There may be minor changes to the UK machines, but they are not expected to be significant if indeed the machine is to be sold in the UK.

MSX is the new Microsoft standard designed to bring compatibility to software in low-priced home personal computers. It is based on computers built around the Z-80A processor, the TMS-9918A video display processor and the AY-3-8910 programmable sound generator. A 32K ROM contains an enhanced version of Microsoft's Basic

interpreter and there is a minimum requirement for 8K of user-RAM.

Companies supporting MSX beside Spectravideo include most of the major Japanese electronics companies who at present do not have a foot in the computing door in this country — Sony, Sanyo, Toshiba, Hitachi, to name but a few. If they are serious then the next few years in computing is going to be very interesting.

One thing the standard will do, if it takes off, is to allow new machines to be launched into a market where, as a great deal of software will already exist, the only edge will be price and styling.

Full-size keyboard

The full-size keyboard with space-bar is of the Spectrum moving-key type and is one of the best examples of its kind with a switchable key click audible through the television speaker. The joystick provides a rapid means of moving the cursor around the screen and can be operated as a touch pad when the central lever is removed, a nice touch.

The cassette interface is at either 1200 or 2400 BPS but was untested. Expansion is through a 50-way edge connector and power input is via a separate transformer unit which provides both a 14V and 8V a.c. supply. Error messages are given in full with the line number of the occurrence. The machine boasts a full screen editor which with the touch pad facility is extremely easy to use.

When running a program which stops because of an error in the Basic, pressing the F4 key brings the offending line onto the screen for correction.

At switch on, there are 12,816 bytes free on this 32K machine. 16K is allocated to the video display processor and the user has the other half. One of the more interesting facilities is the ability to work in integer, single precision and double precision numbers. The default is double precision and that must be useful for business application programs — see figure 1.

Run two programs at once

Another possibly useful facility is the ability to run two programs at the same time — banked program switching — when the optional memory expansion is installed. Quite how this would be used I am not sure as I would not have thought the machine suitable for multi-user applications.

The VDP is capable of displaying 16 colours in high-resolution mode and supporting 32 sprites. The sprites may be single or quad square entities with the capability of being located in different planes to enable one sprite to pass over another. There is a graphics

macro language — GML — which looks to have interesting capabilities in three-dimensional movements. At 256 by 192 the pixel resolution is not high for the first of a new generation of computers, without expansion the text screen display is limited to 40 columns.

The programmable sound generator — PSG — operates separately from the main processor enabling background music to accompany the landing of your flying saucers. The PSG also has its own music macro language — MML. Unfortunately, not being musically oriented, I am not able to use the full potential of the facility but it looks quite good.

Two manuals supplied

The implementation of a joystick on the keyboard seems to be fashionable but of dubious value. Allowing the stick itself to be removed and leaving what is effectively an eight direction cursor pad is very neat.

Two manuals are supplied, the *SV318 Personal Computer Users Manual* and the *Basic Quick Reference Guide*. The manuals are retained in a three ring binder.

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ARITHMETIC FUNCTIONS					
ABS	ATN	COBL	CINT	COS	CSNG
EXP	FIX	FRE	INT	LOG	RND
SGN	SIN	SQR	TAN		

The usual arithmetic operators are implemented plus EQU and IMP.

STRING FUNCTIONS					
ASC	CHR\$	FRE	HEX\$	INKEY\$	
INSTR	LEFT\$	LEN	MID\$	INPUT\$	
OCT\$	RIGHT\$	VAL	SPACE\$	STR\$	STRING\$

I/O & SPECIAL FUNCTIONS					
CUI	CUS	CUD	ERL	ERR	INP
LPOS	MKI\$	MKS\$	MDK\$	PEEK	POS
SPC	TAB	USR	VARPTR	ATN	

BASIC COMMANDS			
AUTO	BLOAD	BSAVE	CLEAR
CLOAD	COLOR	CONT	CSAVE
DELETE	KEY LIST	LIST	LLIST
LOAD	MAXFILES	MERGE	MOTOR
NEW	RENUM	RUN	SAVE
SOUND	SWITCH	SWITCH	STOP
TRON	TROFF	WIDTH	

BASIC STATEMENTS			
BEEP	CLICK	DEF FN	DEF USR
DEFINT	DEFSTR	DEFSNG	DEFDBL
DIM	END	ERASE	ERROR
FOR	GOSUB	GOTO	
IF-THEN-ELSE	IF-GOTO-ELSE		
LET	MID\$	NEXT	ON GOSUB
ON GOTO OUT	POKE	REM	
RESTORE	RETURN	STOP	SWAP
WAIT	SOUND	SWITCH	

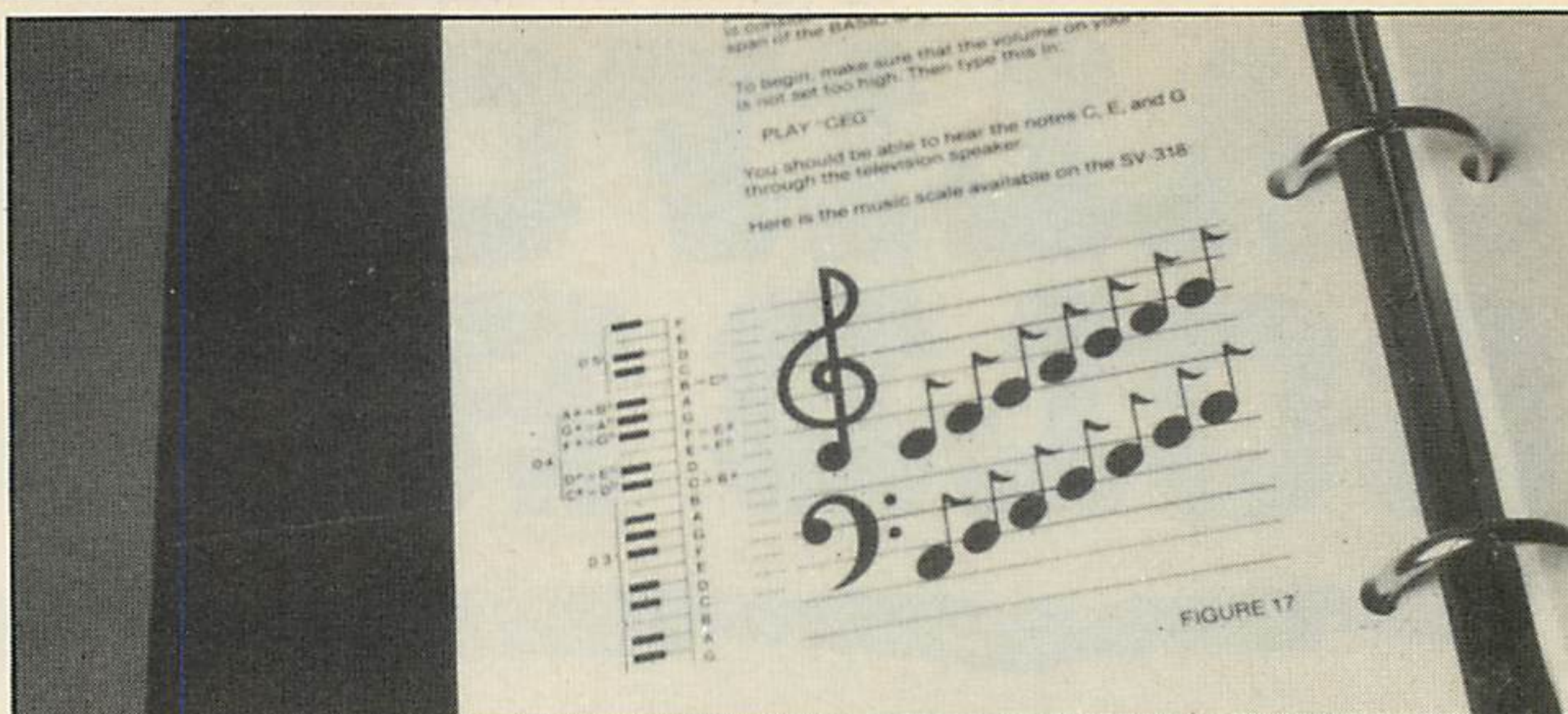
GRAPHIC STATEMENTS			
CLS	CIRCLE	COLOR	DRAW
GET	LINE	LOCATE	PAINT
POINT	PSET	PRESET	PUT
PUT SPRITE	SCREEN	SPRITE\$	
UPEEK	UPOKE		

INTERRUPT CONTROL			
ON ERROR GOSUB			
ON INTERVAL	ON KEY		
ON STOP GOSUB	ON SPRITE GOSUB		
ON STRIG GOSUB			
INTERVAL	KEY	STOP	
STRIG	SPRITE	on/off/stop	



Figure 1. One of the more interesting facilities of the SpectraVideo 318 is its ability to operate in integer, single precision and the default of double precision, eminently suitable for business application programs. As a consequence of using double precision and storing numbers in 8 bytes, this program uses 8049 bytes for variable storage.

```
2DIMB(1000)
3FORA=1TO1000
4B(1001-A)=A
5NEXT
6GOSUB20:PRINT
7C=0
8FORA=1TO99
9IFB(A)>B(A+1)THENGOSUB30
10NEXT
11IFC<>0THENGOTO7
12GOSUB20
13STOP
20FORA=1TO100
21PRINTB(A);
22NEXT
23RETURN
30D=B(A)
31B(A)=B(A+1)
32B(A+1)=D
33C=1
34RETURN
```



(continued from page 77)

The user manual falls some way short of teaching the beginner how to use the machine but there is a tutorial tape which I have not seen and also a Basic Reference Guide and I assume that these items will complement the material in the manuals in that respect.

The memory maps in the manual interestingly defines Bank 2 as being reserved for CPM.

Prices.	
SPECTRAVIDEO 318	£199
CASSETTES	£6
GAMES CARTRIDGES	£16
16K RAM	£29
64K RAM	£89
MINI EXPANDER	£12
7SLOT MOTHERBOARD	£99
DISC DRIVES	£199
SPECTRAVIDEO 328	£250
Real keyboard and 80K RAM.	

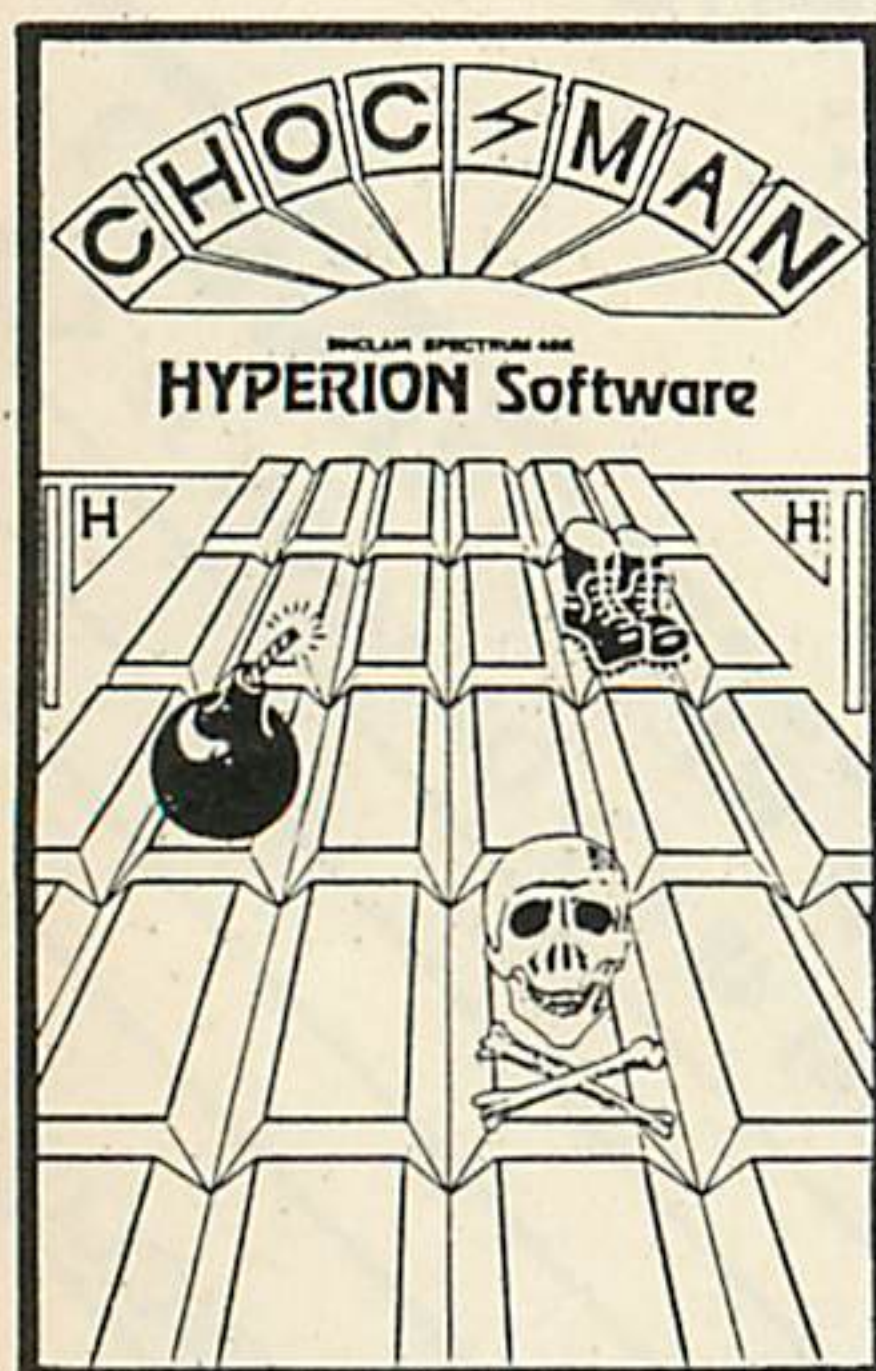
CONCLUSIONS

- In this short appraisal I have only highlighted the areas that seem to me to be of note. In doing so, much of the usual content of a review has been omitted.
- The machine is quick with some very useful sound and graphic capabilities. It costs about \$300 in the United States which translates to £200 in the U.K.
- It could hold its own against all the current machines in that price bracket, but the new 8-bit machines may well cause it problems.
- Everything will depend upon the pricing levels adopted and how determined the MSX club is to penetrate the U.K. market.

SUPER SPECTRUM GAMES

CHOC-MAN

Move around the blocks to defuse the ticking bombs but beware of the skulls and stomping boots, flags give extra points. Four stages to get through. 100% machine code for the 48K Spectrum. Only £5.95

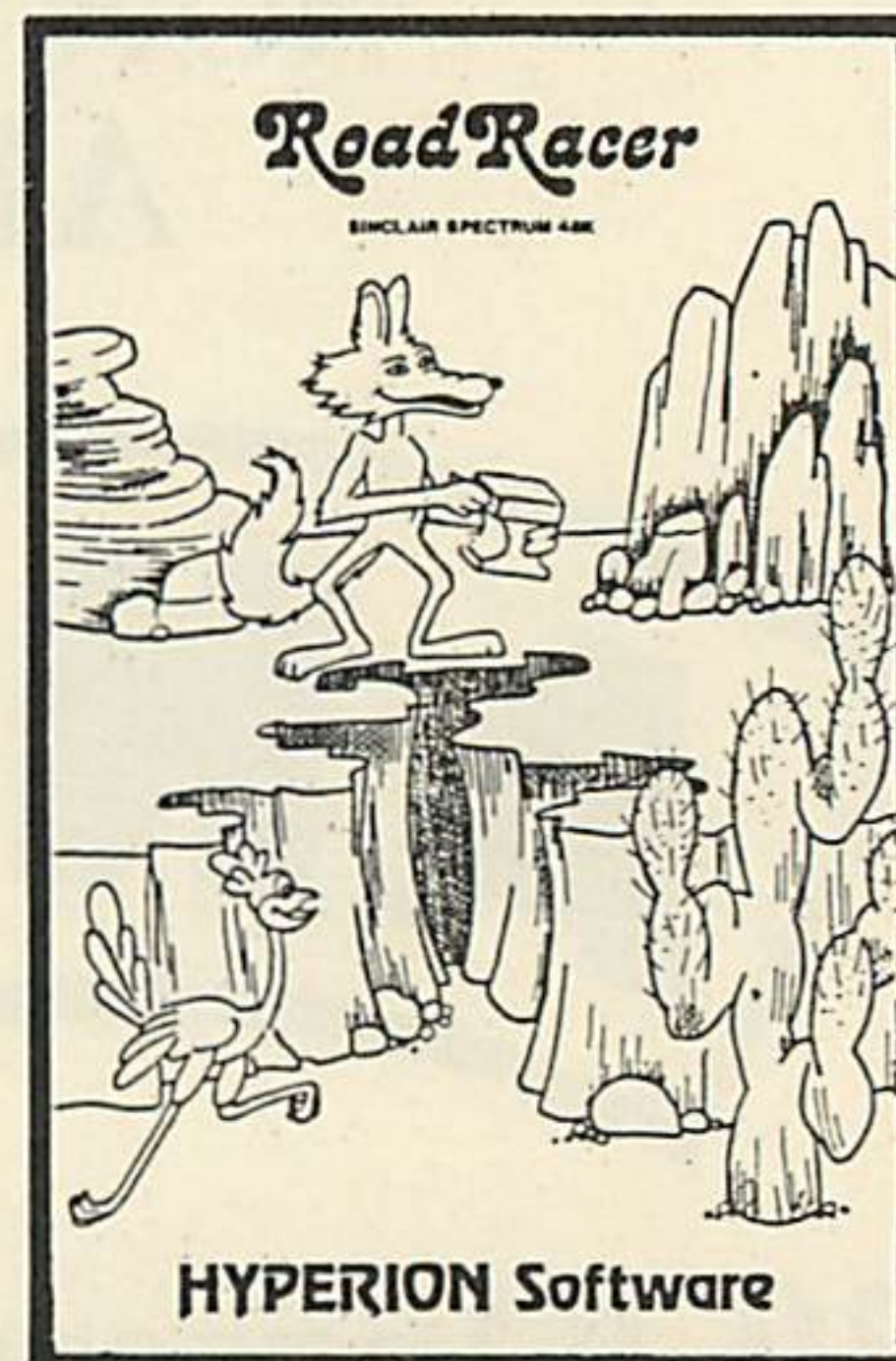


SPIDERON

(One game your mother and sister will not want to play) can you climb to the top of a pit with giant spiders attacking from everywhere? Can you? Dare you? Its a fast and very infuriating game. 100% machine code for 16 or 48K Spectrum. Only £5.95

ROADRACER

Its time that coyote caught that there chicken. Its an easy to play game, but its sure hard to beat. There are four screens to get through, but one mistake, and you go all the way back to the start. Written for the 48K Spectrum. Only £5.95



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145 HANOVER ST
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